

labbook demo

As of commit 1e92402

labbook demo

2026-09-25

Table of contents

1	Overview	3
1.1	Entries	3
1.2	Open events (0)	3
1.3	Latest results	3
2	Conventions	4
2.1	Identifiers and labels	4
2.2	Numbers	4
2.3	Project conventions	4
I	September 2026	5
3	Week 39 · 25 Sep 2026	6
3.1	Friday, 25 September 2026	6
3.1.1	Session Convergence of explicit Euler: Plan	6
3.1.2	Order of convergence of explicit Euler	7
3.1.3	Does step halving still work for coarse steps?	9
3.1.4	Session Convergence of explicit Euler: Summary	11

1 Overview

This lab notebook is generated from Markdown sources (Quarto). The sources of truth are the `.qmd` files, `results.tsv`, `runs/*/provenance.json` and the mechanical trace in `_trace/`. This PDF/HTML is derived and is never edited by hand.

1.1 Entries

Entry	Opened	Closed	Status	Verdict	Runs	Review
Order of convergence of explicit Euler	2026-09-25 10:18	2026-09-25 10:18	closed	confirmed	R-0001, R-0002, R-0003	–
Does step halving still work for coarse steps?	2026-09-25 10:18	2026-09-25 10:18	closed	refuted	R-0004, R-0005	–

1.2 Open events (0)

None.

1.3 Latest results

Run	Metric	Value	Reference	Status
R-0005	observed_order	1.19542	1	keep
R-0005	error_ratio	2.29011	2	keep
R-0005	abs_error	0.0514732		keep
R-0004	abs_error	0.117879		keep
R-0003	observed_order	1.00606	1	keep
R-0003	error_ratio	2.00842	2	keep
R-0002	observed_order	1.01224	1	keep
R-0002	error_ratio	2.01704	2	keep
R-0003	abs_error	0.0018471		keep
R-0002	abs_error	0.00370976		keep
R-0001	abs_error	0.00748272		keep

2 Conventions

Maintainer: adapt this file to the project, then run `python3 tools/lb.py protect` and commit. It is write-protected for the assistant: conventions are set by the human.

The conventions below bind every entry. An entry that deviates from one says so explicitly.

2.1 Identifiers and labels

- Entries: YYYY-MM-DD_slug; runs: R-NNNN; events: E-NNNN.
- Quarto labels carry the entry slug as prefix: `fig-<slug>-<name>`, `eq-<slug>-<name>`, `tbl-<slug>-<name>`; cross-references with `@fig-<slug>-<name>`.

2.2 Numbers

- Numbers in the text come from the ledger via shortcodes (`{{< result R-0007 metric >}}`, `{{< run-param R-0007 key >}}`); a number typed by hand names its source.
- Every quantity carries its unit.

2.3 Project conventions

Part I

September 2026

3 Week 39 · 25 Sep 2026

3.1 Friday, 25 September 2026

3.1.1 Session Convergence of explicit Euler: Plan

2026-09-25_euler-study/plan.qmd · start 2026-09-25 10:18

3.1.1.1 Goal

Establish the order of convergence of the explicit Euler integrator in `src/decay.py` and the range of step sizes in which the textbook first-order behaviour holds.

3.1.1.2 Hypotheses

1. **H1** For $h \leq 0.04$ s the error at $t = 1$ s halves when h halves: every error ratio lies in $[1.9, 2.1]$.
2. **H2** The same holds for the coarse pair $h = 0.5 \rightarrow 0.25$ s.

3.1.1.3 Planned experiments and tests

Runs at $h = 0.04, 0.02, 0.01$ s (H1) and $0.5, 0.25$ s (H2); each takes milliseconds.

3.1.1.4 Scope of action

May change: the notebook and the analysis scripts. May not change: `src/decay.py`, the decision interval.

3.1.1.5 Stop criteria

Stop after the five runs. Further step sizes only with a new plan.

3.1.1.6 Test oracle

The exact solution $y(t) = e^{-kt}$ with $k = 1 \text{ s}^{-1}$.

3.1.2 Order of convergence of explicit Euler

2026-09-25_euler-order · opened 2026-09-25 10:18 · closed 2026-09-25 10:18 · status closed · verdict **confirmed** · runs R-0001, R-0002, R-0003

3.1.2.1 Why

The integrator `src/decay.py` uses the explicit Euler method, which is expected to be first-order accurate. Before it is used for longer integrations, its order is checked on a problem with a known solution (plan H1).

3.1.2.2 Hypothesis / Expectation

For $dy/dt = -ky$, $y(0) = 1$, $k = 1\text{ s}^{-1}$, the absolute error at $t = 1\text{ s}$ halves when the step halves: both error ratios, for $h = 0.04 \rightarrow 0.02\text{ s}$ and $0.02 \rightarrow 0.01\text{ s}$, lie in $[1.9, 2.1]$. Refuted if either ratio falls outside.

3.1.2.3 Method

Sources

`src/decay.py`; the analysis script `analysis.py` of this entry.

Equations and code references

$$y_{n+1} = y_n - h k y_n \quad (3.1)$$

implemented in `src/decay.py:14-19`; h step size in s, k decay rate in s^{-1} , y dimensionless. Error and observed order (`analysis.py`, functions `final_error` and `main`):

$$\varepsilon(h) = |y_N - e^{-kt_{\text{end}}}|, \quad p = \frac{\ln[\varepsilon(h_1)/\varepsilon(h_2)]}{\ln(h_1/h_2)} \quad (3.2)$$

Parameters and test coverage

Run	h [s]	Purpose
R-0001	0.04	coarsest of the fine steps
R-0002	0.02	
R-0003	0.01	finest

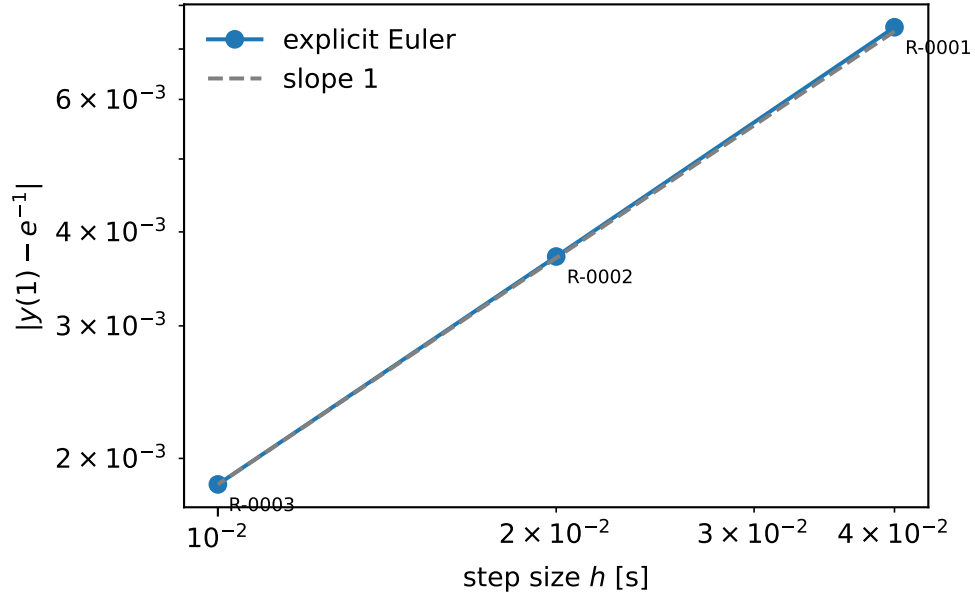


Figure 3.1: Absolute error at $t = 1$ s against the step size, with a slope-1 reference line. *Run R-0003, commit db53137, 2026-09-25*

3.1.2.4 Result

Fig. 3.1 shows the error falling with the step size. The error ratios are 2.01704 (R-0001 $\rightarrow$ R-0002) and 2.00842 (R-0002 $\rightarrow$ R-0003); the observed orders (Eq. 3.2) are 1.01224 and 1.00606.

3.1.2.5 Verification

Oracle: the exact solution e^{-t} . Limiting case: the error decreases monotonically towards zero with h in all three runs. Conservation laws: not applicable (one linear equation).

3.1.2.6 Interpretation

Both ratios lie inside the decision interval. The observed order approaches 1 from above as h decreases, as expected when an $O(h)$ leading error carries an $O(h^2)$ correction.

3.1.2.7 Consequences and next steps

Only $hk \leq 0.04$ was tested. Whether the rule also holds for coarse steps is plan hypothesis H2 (next entry).

3.1.2.8 Failed attempts

none

3.1.2.9 Deviations from plan

none

3.1.3 Does step halving still work for coarse steps?

*2026-09-25_coarse-steps · opened 2026-09-25 10:18 · closed 2026-09-25 10:18 · status closed · verdict **refuted** · runs R-0004, R-0005*

3.1.3.1 Why

The previous entry confirmed first order for $hk \leq 0.04$. Coarse steps are tempting for long integrations; plan hypothesis H2 asks whether step halving still predicts the error there.

3.1.3.2 Hypothesis / Expectation

The error ratio for $h = 0.5 \rightarrow 0.25$ s lies in $[1.9, 2.1]$, as for fine steps. (The author expects a refutation, because $hk = 0.5$ is far from the asymptotic regime; the interval decides.)

3.1.3.3 Method

Sources

Same model and analysis as the previous entry.

Equations and code references

As in the previous entry: the Euler step in `src/decay.py:14-19`, error and observed order in `analysis.py`.

Parameters and test coverage

Run	h [s]	Purpose
R-0004	0.5	coarse
R-0005	0.25	half of it

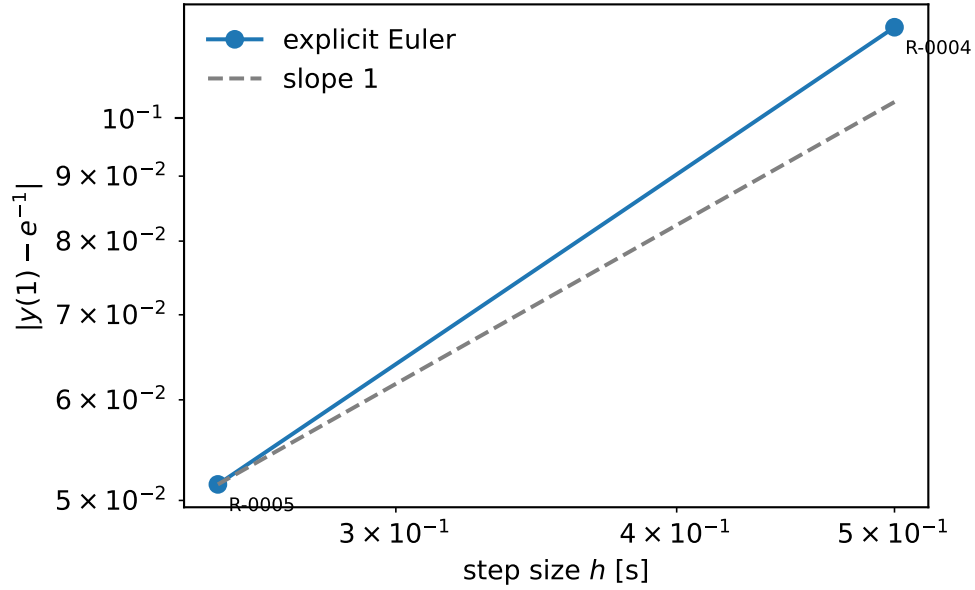


Figure 3.2: Absolute error at $t = 1$ s for the coarse steps, with a slope-1 reference line. *Run R-0005, commit bc8732f, 2026-09-25*

3.1.3.4 Result

The error ratio is 2.29011, outside the decision interval; the observed order is 1.19542 (Fig. 3.2).

3.1.3.5 Verification

Same oracle as before; both errors are positive and decrease with h .

3.1.3.6 Interpretation

At $hk = 0.5$ the $O(h^2)$ terms are not negligible and the method has not yet reached its asymptotic regime. First order is a statement about $h \rightarrow 0$, not a rule for coarse steps.

3.1.3.7 Consequences and next steps

Error estimates by step halving need $hk \lesssim 0.04$ for this problem. The boundary between $hk = 0.04$ and 0.5 was not resolved.

3.1.3.8 Failed attempts

none

3.1.3.9 Deviations from plan

none

3.1.4 Session Convergence of explicit Euler: Summary

2026-09-25_euler-study/summary.qmd · start 2026-09-25 10:18 · end 2026-09-25 10:18

3.1.4.1 Planned

Two hypotheses on the order of explicit Euler (plan H1, H2).

3.1.4.2 Carried out

Five runs in two entries: fine steps (R-0001 to R-0003) and coarse steps (R-0004, R-0005). This demo ran without Claude Code, so the mechanical trace below has no records.

3.1.4.3 Deviations from plan

none

3.1.4.4 Key findings

H1 confirmed: observed order 1.00606 at the finest pair. H2 refuted: error ratio 2.29011 for the coarse pair.

3.1.4.5 Open questions and next steps

Where between $hk = 0.04$ and 0.5 the asymptotic regime begins.

3.1.4.6 Appendix: Trace

Trace statistics (generated mechanically)

Period: – to –; Claude sessions: unknown; trace records: 0

Tool	Calls	of which failed
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Modified file	Changes
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Shell commands: 0, of which failed: 0.

Event	Kind	Detail	Documented
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